

COAL Assignment # 2 (Submission Date: 30th October, 2022, 11:59pm)

Question#1: Write a subroutine that takes two parameters 1) a 16-bit number and 2) an n-bit ($n < 16$) pattern. Now find whether the given n-bits can be found in 16-bit number or not. If n-bits pattern is found, then return the starting index number else return -1. Suppose that the index number of the most significant bit is 0, and the index number of the least significant bit is 15. This sub routine will return starting Index number if found else -1.

For example:

Example 1: <i>AX has 16 bit number and BX has n bit pattern</i> <i>AX= 1110 1111 0000 1010b</i> <i>BX= 111100b (it does not matter what the remaining 9 bits of BX contain)</i> <i>n=6</i> <i>In example 1, the n-bit number can be found in AX, starting from index=4</i>	Example 2: <i>AX= 1100 1100 1010 0000b</i> <i>BX= 1101b (it does not matter what the remaining 12 bits of BX contain) n=4</i> <i>In example 2, the given n-bit number cannot be found in AX.</i>
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Question#2: Write a function named statsOfArray that takes an array as input and returns its min, max and median as output. All the input and output parameters should be passed on stack. You cannot sort the array in statsOfArray function, however, you can call BubbleSort function in statsOfArray. Template of this function is given below, your code should work for every array.

```
[org 0x0100] jmp
start
array1: dw 6, 7, 3, 9, 100, 5, 1, 50
lenArray1: dw 8
statOfArray:
; this function will take one array as input
; and will return min, max and median of array.
; median is the middle element of sorted array
; write your function here start:
; make space for o/p ;
push inputs for array1
call statOfArray
pop ax ; ax should be min of array
pop bx ; ax should be max of array
pop cx ; ax should be median of array
mov ax, 0x4c00
int 21h
```

Question# 3: Write a function structure that takes 4 input variables, returns 3 outputs and creates 5 local variables, function only used ax, bx and si, di, cx therefore save and restore these registers. Pop the outputs in ax, bx and cx after return. Create its stack and specify how LV, I/P and O/P will be accessed inside subroutine